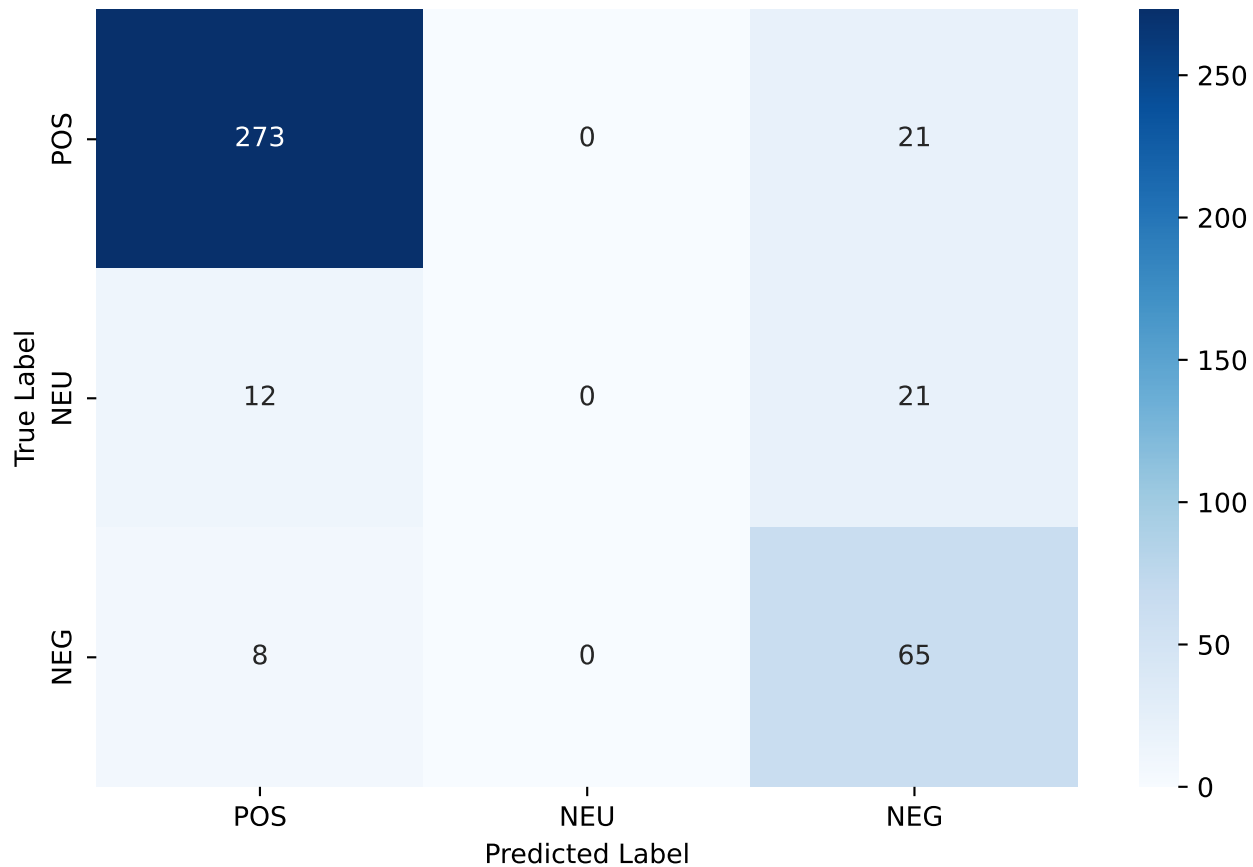
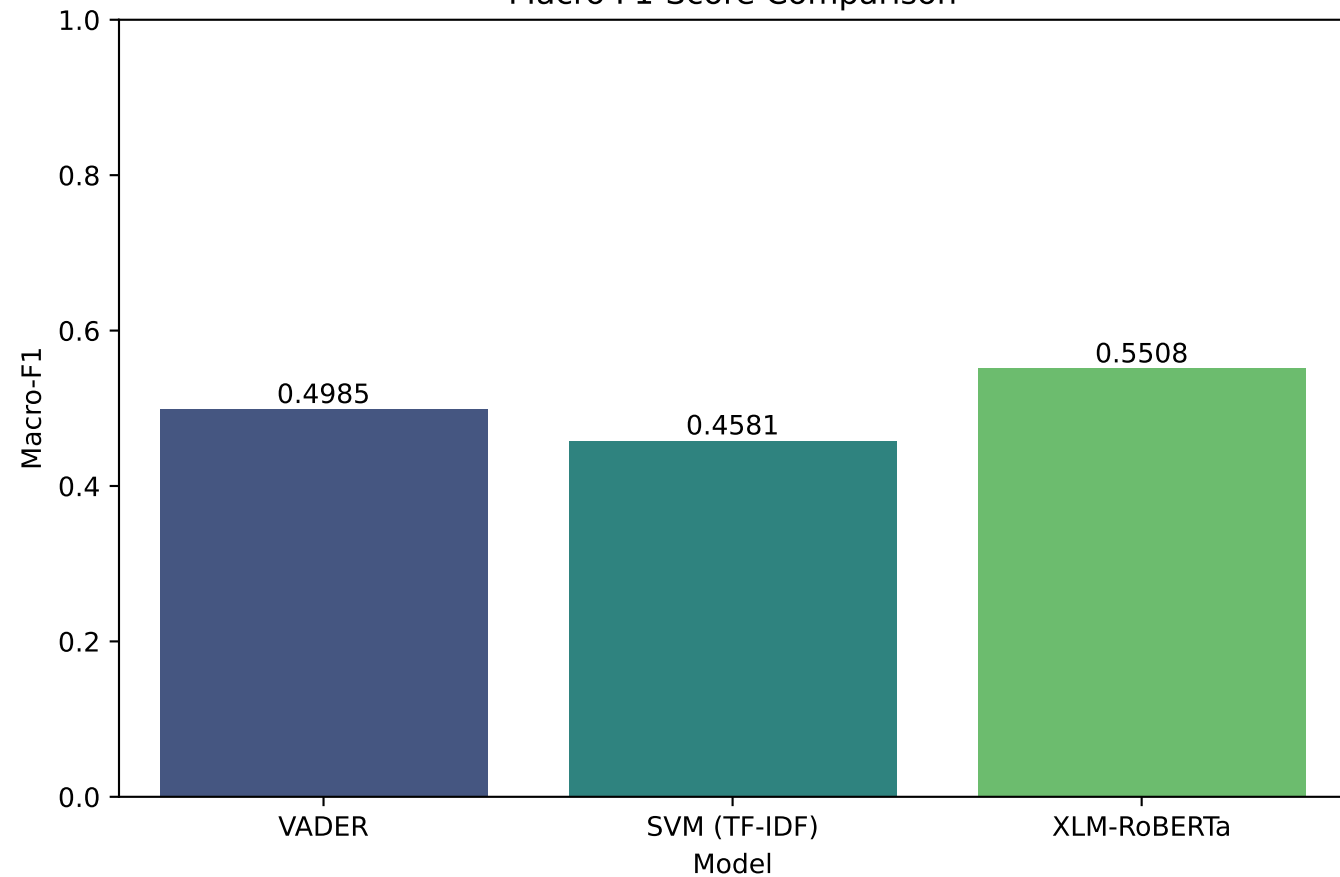


XLM-RoBERTa Confusion Matrix (Phase 8)



Macro F1-Score Comparison



PHASE 8: OUT-OF-SAMPLE EVALUATION REPORT

1. FINAL BENCHMARK METRICS (Expanded)

Model	Accuracy	Precision (Macro)	Recall (Macro)	Macro-F1	F1 (POS)	F1 (NEU)	F1 (NEG)
VADER	0.6875	0.5685	0.5052	0.4985	0.8293	0.2034	0.4630
SVM (TF-IDF)	0.7675	0.4427	0.4751	0.4581	0.8710	0.0000	0.5034
XLNet-RoBERTa	0.8450	0.5131	0.6063	0.5508	0.9302	0.0000	0.7222

2. MCNEMAR'S STATISTICAL TEST (alpha=0.05)

Baseline	XLNet-R	Correct (b)	Base Correct (c)	Chi-Square	p-value
VADER Lexicon		85	22	37.0935	1.1260e-09
SVM (TF-IDF)		44	13	16.8596	4.0248e-05

3. PHASE 7 GRID SEARCH SUMMARY

Batch Size	Learning Rate	Epochs	Status
16	2e-5	5	Suboptimal
16	3e-5	5	Suboptimal
16	5e-5	5	Suboptimal
32	2e-5	5	Suboptimal
32	3e-5	5	Optimal
32	5e-5	5	Suboptimal